Prodigy Anesthesia Quick Review

Forced Air Warming Blanket

What major organ produces the greatest amount of heat as a byproduct of its metabolic function?	All of the major organs produce a significant amount of heat, but the brain produces the most. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1505.*
What are the three components to a forced air warmer?	An inflatable blanket, an electrical control unit, and a hose *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 265.*
Why should heavy blankets not be placed over a forced-air blanket?	Heavy blankets force all of the warmed air directly to the patient's skin, increasing the risk for burns. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 301.*
Why should forced air warmers not be applied to ischemic areas or areas with poor perfusion?	There is a risk for burns *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 265.*
What method of heat transfer do forced air warmers use?	Convection *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 265.*
What is the definition of hypothermia?	Hypothermia is said to exist when the core body temperature is less than 96.8 degrees F or 36 degrees Celsius. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1505.*
What is the central 'controller' for body temperature?	The anterior hypothalamus *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1506.*
What is nonshivering thermogenesis?	Nonshivering thermogenesis is a norepinephrine-mediated mechanism of heat production that involves uncoupling of oxidative phosphorylation within the mitochondria-rich brown fat cells found over the neck, back, viscera, and great vessels. Nonshivering thermogenesis occurs in premature infants and full-term newborns, but does not occur in adults. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1507.*

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What is does the term 'interthreshold range' mean, and how does anesthesia play a role in it?	The interthreshold range is the range of body temperature at which the compensatory responses for cold are not activated and the compensatory mechanisms for heat are not activated either. In a normal, healthy adult, this range is about 0.4 degrees Celsius. Anesthesia widens this range to about 7.2 degrees F or 4 degrees C. This means that ambient temperature and heat loss mechanisms play a larger part in determining core body temperature. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1507.*
The loss of core temperature that occurs under anesthesia follows a predictable pattern. Describe this pattern of heat loss.	There is an initial rapid decrease in core temperature by about 0.9-2.7 degrees F (0.5-1.5 degrees Celsius) during the first hour of anesthesia. A gradual decline occurs that lasts about 2-3 hours. A plateau occurs at about 3-4 hours where heat loss and heat production are equal and the temperature no longer changes. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1507.*
What is the first compensatory response to heat loss?	Cutaneous vasoconstriction, which can decrease heat loss by 25-50% *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1506.*
When does the body initiate shivering as a means of increasing the core body temperature?	Shivering is only initiated after cutaneous vasoconstriction and nonshivering thermogenesis fail to maintain core body temperature. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1506.*
What is the principal disadvantage of shivering?	Shivering increases total body oxygen consumption by 2-3 times. Shivering also increases intraocular and intracranial pressure. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1506.*
What composes the core compartment versus the peripheral compartment?	The core compartment is considered to be composed of the trunk, the head, and the major organs. The peripheral compartment consists of the skin, the superficial tissues of the trunk, the arms, and the legs. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1505-1506.*
According to U.S. standards, what is the AVERAGE contact surface temperature a forced-air device cannot exceed?	46 degrees Celsius *Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 887.*

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According to U.S. standards, what is the MAXIMUM contact surface temperature a forced-air device cannot exceed?

48 degrees Celsius

Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 887.

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